

Results and Validation

Dataset-Free Inverse Design of Electrostatic MEMS via the Pull-In Bifurcation

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TABLE I

Device and simulation parameters. Source: `sim/inverse_design.py`,
`sim/beam.py`.

Parameter	Sym.	Value	Parameter	Sym.	Value
Beam length	l	1000 μm	Nominal gap	d_0	1.0 μm
Beam width	w	10 μm	Travel spec.	T	2.0 μm
Beam thickness	h	100 μm	Fringing par.	α	0.0042
Young's modulus	E	169 GPa	Residual stress	N_t	0
Poisson ratio	ν	0.32	Design grid	N	60
Plane-strain mod.	$\bar{E}$	188.28 GPa	Basis modes	K	12
Permittivity	ε_0	8.854 pF/m	Voltage scale	V_s	1.8826 V

TABLE II

Grid convergence, cantilever, uniform gap, $\alpha = 0$. Deviation is
measured against the finest grid. Source: `sim/validate_beam.py`.

N	Λ_{PI}	Dev. vs $N=120$	$\ R\ $
30	1.67719	-0.203%	7×10^{-10}
45	1.67922	-0.083%	7×10^{-9}
60	1.67993	-0.041%	2×10^{-8}
90	1.68043	-0.011%	1×10^{-7}
120	1.68061	—	4×10^{-7}

Scope. This document supplements the technical report with the full quantitative record: device parameters, numerical verification, validation against eight published sources, and controlled comparisons against a published hand-tuned study and against machine-learning surrogates at equal budget. Every value is produced by a named script and regenerable from a clean checkout.

Run every result yourself. No install, licence, GPU or dataset; each notebook clones the repository in its first cell.

Repository (MIT):

<https://github.com/thetushardudeja1/mems-differentiable-pullin>

Notebook 1: Validation and results (~ 4 min):

https://colab.research.google.com/github/thetushardudeja1/mems-differentiable-pullin/blob/main/notebooks/ME_MS_differentiable_pullin.ipynb

Notebook 2: How the method works (~ 3 min):

https://colab.research.google.com/github/thetushardudeja1/mems-differentiable-pullin/blob/main/notebooks/method_deep_dive.ipynb

I. Device and Simulation Parameters

Table I fixes the benchmark. The governing equation, in nondimensional form with $\xi = x/l$, $Y = y/d_0$ and gap $G = D(\xi) - Y$, is

$$Y'''' - N_t Y'' = \Lambda \frac{1 + \alpha G}{G^2}, \quad \Lambda = \frac{6\varepsilon_0 V^2 l^4}{\bar{E} w^3 d_0^3}. \quad (1)$$

Terms used throughout. Travel is the stable displacement the beam can reach before pull-in; it is the useful stroke of the actuator, and 2 μm is the specification held fixed here so that every design is compared at equal capability. Tip-in is the competing failure mode of a 2-DOF actuator, in which the beam rotates into contact at one end before it snaps down flat; the angle at which it

occurs is set by a fabrication parameter β that cannot be measured without destroying the device. Λ_{PI} is the dimensionless pull-in load of (1), so $V_{\text{PI}} \propto \sqrt{\Lambda_{\text{PI}}}$. A true solve is one full Newton solution of (2), as opposed to a surrogate evaluation; an amortized network is one trained once so that it can emit a design for any specification without optimising again; headroom is the gap between what a worst-case fixed controller achieves and what an oracle with perfect knowledge of each device could achieve.

Every voltage in this document is computed on the same $N = 60$ grid with every design re-fitted to exactly 2.000 μm of stable travel. Comparing designs that do not meet the same specification would be meaningless, and comparing designs fitted on different grids has flipped the sign of a conclusion in this project before.

Hardware and precision. All timings are single-threaded on an Intel-class laptop CPU; the solver runs in float64 and converges to $\|R\| \sim 10^{-8}$, so voltages are repeatable to better than 10^{-3} V run to run and the digits quoted below are meaningful.

II. Numerical Verification

Grid. Table II shows Λ_{PI} changing by 0.041% between $N = 60$ and the finest grid tested. $N = 60$ is therefore converged to well inside every discrepancy reported below, and is used throughout.

The fold is solved, not sampled. Under voltage control the equilibrium branch turns around at pull-in and $\partial Y / \partial \Lambda \rightarrow \infty$. We instead solve the extended fold system

$$R(Y, \Lambda) = 0, \quad J(Y, \Lambda) v = 0, \quad \|v\| = 1 \quad (2)$$

for (Y, Λ, v) simultaneously [6], locating the fold as the solution of an equation rather than the maximum of a sampled curve (Fig. 1). Residuals are $\sim 10^{-8}$ throughout.

The gradient is exact. Because (2) is a smooth root-finding problem, the implicit function theorem returns

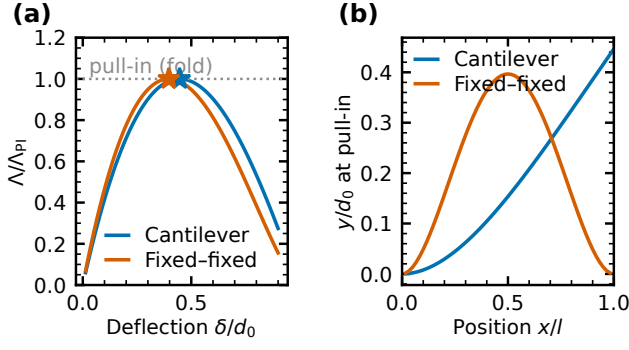


Fig. 1. The fold. Under displacement control the branch is smooth and single-valued through pull-in, where the Jacobian is singular; the deflection shape at the fold is the null vector of (2).

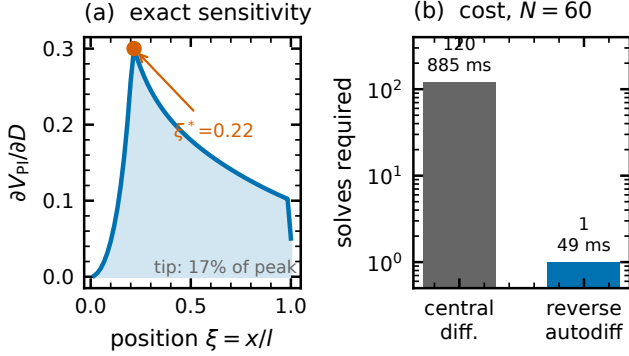


Fig. 2. (a) The exact sensitivity $\partial V_{PI}/\partial D(\xi)$. It peaks near the clamp at $\xi^* = 0.22$, not at the free tip where deflection is largest. (b) One reverse pass replaces $2N$ forward solves.

$\partial V_{PI}/\partial d(x)$ at every point from one reverse pass. With fringing off, Λ_{PI} is homogeneous of degree three in the gap profile, so Euler’s homogeneous-function theorem requires $\sum_i D_i \partial \Lambda / \partial D_i = 3\Lambda$ with no adjustable constant. Measured: 0.000000% error. An approximate gradient cannot satisfy this.

Cost, and where the leverage is (Fig. 2). Central differences need $2N$ solves for an N -node sensitivity; the reverse pass needs one. At $N = 60$ that is 120 solves against one, measured $18\times$ faster, and the ratio grows with N . The two agree to 2.4×10^{-5} relative at the dominant node, which is the finite-difference truncation error rather than the autodiff.

The curve itself is a design result. Sensitivity peaks at $\xi^* = 0.217$, near the clamp, and the free tip carries only 16.7% of it. The electrode shaping that buys drive voltage therefore happens in the first quarter of the beam, which is the part a designer is least likely to tune by hand (sim/sensitivity_map.py, sim/make_fig_sensitivity.py).

III. Validation Against Eight Sources

Table III spans 1967 to 2025 and includes two physically measured devices. Two entries carry more weight than the rest.

TABLE III
Validation against eight independent sources, 1967–2025. The last two rows are measured devices. † Normalised, as in [5].

Source	Quantity	Ours	Ref.	Error
Closed form	λ^* , lumped	0.148150	0.148148	0.001%
Nathanson [9]	Λ_{PI} cant.	1.6799	1.6790	0.05%
Nathanson [9]	Λ_{PI} fix-fix	69.935	71.167	1.73%
Analytic c^3	$\sum \partial \Lambda / \partial D$	5.0398	5.0398	0.00%
Seeger [4]	tip-in, design #1	19.0%	19%	exact
Seeger [4]	tip-in, design #3	85.1%	85%	0.12%
Seeger [4]	mechanism, 3 designs	3/3	3/3	—
Nazemi [5]	V_{PI} reduction †	13.2%	16.0%	2.8 pp
M-TEST [2]	V_{PI} , 500 μm	11.70 V	11.1 ± 0.1 V	5.4%

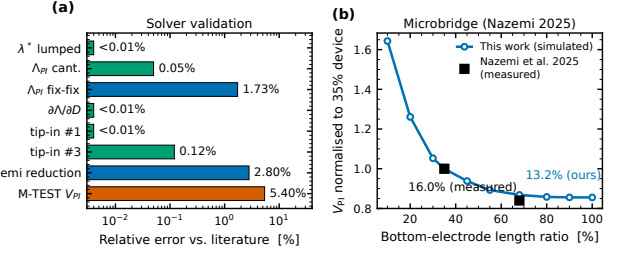


Fig. 3. Error across all validation sources, and the 2025 measured microbridge [5] against our prediction.

The analytic c^3 law is a gradient check with no fitted constant, as above. The Seeger and Boser entry is a mechanism prediction rather than a fitted number: their designs #3 and #4 are mechanically identical and differ only in parasitic capacitance, and the model correctly predicts that this single electrical change flips the failure mode from tip-in to charge pull-in and costs 20% of travel. All three mechanisms are predicted correctly.

The M-TEST comparison also isolates a physical effect: setting residual stress to zero moves the predicted V_{PI} from 11.70 V to 10.16 V, so tension supplies 18% of the predicted voltage on that device. It is not a small correction, and the model resolves it.

On the fixed–fixed entry. Its 1.73% is the largest deviation in the table, against 0.05% for the cantilever, and the reason is the reference rather than the solver: the fixed–fixed closed form is itself an approximation fitted to a lumped mode shape, whereas the cantilever expression is exact in the same limit. Our value is grid-converged to 0.04% (Table II), and the design conclusions rest on relative comparisons within one boundary condition, where this offset cancels.

IV. Inverse Design vs. a Published Study

Free-form optimisation over 60 degrees of freedom beats the best family Haluzan et al. scanned by hand, in both geometries, without being told the family exists (Table IV).

Independently matched, never fitted. The optimiser recovers $d_{\max} = 6.571 \mu\text{m}$ against their published 6.57 μm and a flattened-bottom ratio of 0.80 against 0.78.

TABLE IV

Inverse design versus [3], both boundary conditions. Every design re-fitted to 2 μm stable travel at $N = 60$. Voltages in V.

Design	Cantilever		Fixed-fixed	
	Ours	Paper	Ours	Paper
Uniform gap	22.86	23.66	175.83	178.26
Linear ($n=1$)	14.31	14.86	123.01	125.48
Best family in [3]	13.61	14.14	121.79	123.94
Free-form (ours)	13.60	—	120.03	—
Optimal exponent n^*	1.29	4/3	1.00	1.00
Gain over best family	0.14%	—	1.45%	—

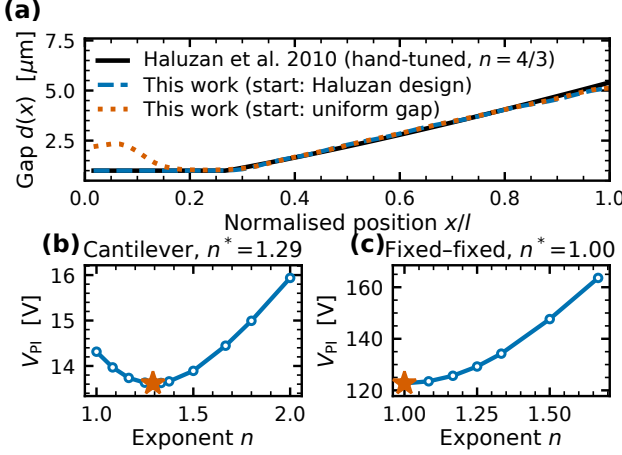


Fig. 4. (a) Both optimisation runs, one started from the published family and one from a plain uniform gap, converge on the same free-form profile (solid), which is the reported design. (b),(c) The optimal exponent reverses between boundary conditions; stars mark n^* .

Positive controls. From a uniform gap, with no knowledge of the published family, the optimiser descends $22.86 \rightarrow 13.87$ V (cantilever) and $175.8 \rightarrow 123.5$ V (fixed-fixed) unaided, converging on the published profile shape (Fig. 4a). The result is not an artefact of initialisation.

An open question, answered. Haluzan et al. conjectured that gains beyond simple shapes “are believed to be minimal.” Measured on their own benchmark: 0.14% and 1.45%. They were right, and it is now quantified rather than believed.

The reversal is the strongest evidence here. The optimal exponent moves from $n^* \approx 4/3$ to $n^* = 1.00$ purely because the boundary condition changed (Fig. 4b,c). A surrogate fitted to cantilever data has no mechanism by which to know this: the reversal is a property of the governing equations, not of the samples.

V. Comparison Against ML Surrogates

The fair question is not whether a surrogate can work, but who ends up with the better device given the same number of true solves. Table V answers it at 400 solves, every design scored by the exact solver, with the surrogate baselines strengthened until they stopped being strawmen.

Direct differentiation reaches 13.783 V in 15 s with zero training samples. Warm-starting from the amortized

TABLE V

All methods on one task (cantilever, 2 μm travel) at an equal budget of 400 true solves, scored by the exact solver. MLP is a multi-layer perceptron and FNO a Fourier Neural Operator [8]; both were given active learning, a trust region and feasible restarts. Times exclude the network’s one-time training (1577 s, 6000 live solves).

Method	Samples	Time	V_{PI} [V]
MLP surrogate, active learning	400	71 s	17.203
FNO surrogate, active learning	400	594 s	14.245
Hand-tuned design [3]	—	manual	14.140
Direct differentiable	0	15 s	13.783
Direct, 4 restarts	0	170 s	13.665
Amortized network	0	0.04 ms	13.636
Network + 100 grad. steps	0	17 s	13.599

TABLE VI

Methodology against the standard dataset-driven pipeline. [1] values are as reported there, for a different task, so this compares method and not performance.

	Zhang [1]	This work
Simulations in dataset	48,000	0
Gradient source	fitted surrogate	exact, through fold
Inference per design	0.01 s	0.04 ms
Change the spec.	re-fit	network input
Change boundary cond.	regenerate data	one argument
Commercial licence	FEA for data	none

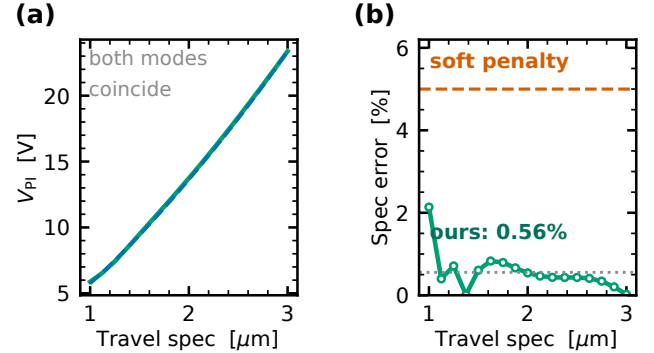


Fig. 5. Specification sweep over the full 1–3 μm range. (a) Feed-forward and enforcement-layer designs give the same drive voltage to within line width, so imposing the constraint costs nothing in V_{PI} . (b) Specification error: 0.56% mean / 2.14% worst case, against $\sim 5\%$ for a soft penalty.

network reaches 13.599 V, better than the published hand-tuned design at 14.140 V and $40\times$ faster in wall-clock than the FNO route.

Table VI places this beside the standard pipeline [1], which generates 48,000 simulations before designing anything. The last two rows are the practical argument: a dataset-fitted model is bound to the design space its samples covered, and changing the boundary condition means regenerating the dataset. Here it costs one function argument, which is exactly what the exponent reversal demands.

TABLE VII
Adaptive policy vs. the best safe fixed-gain target. Three independent seeds $\times$ 512 held-out devices. Source: sim/analyze_rl2dof.py.

Policy	Travel	Destroyed	ρ vs ceiling
Best safe fixed gain	0.5674	0.0%	+0.06
Adaptive policy	0.6399 ± 0.0044	0.0%	+0.99
Oracle (knows β)	0.6577	0.0%	—

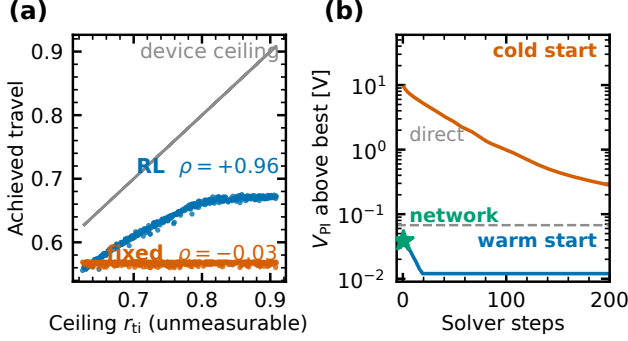


Fig. 6. (a) Achieved travel tracks each device's own hidden ceiling where a fixed gain cannot. (b) Warm-starting from the amortized network reaches a better design in 5 steps than a cold start reaches in 500; the dashed line is converged direct optimisation at 13.66 V.

VI. Amortized Design and Hard Constraints

Optimising one device at a time costs ~ 15 s per specification. A small network trained through the solver removes that: it emits a finished electrode profile from the specification alone in 0.04ms, having seen 0 stored training samples (1577s of training, 6000 live solves).

How the specification is met is the interesting part. A soft penalty on the travel constraint is biased: shrinking travel also shrinks V_{PI} , so the optimiser is rewarded for missing the specification and settles near 5% error. A differentiable enforcement layer instead solves $\text{travel}(p, \sigma) = T$ by Newton and carries the gradient through it by the implicit function theorem [7], cutting error to 0.56% mean and 2.14% worst case across a 17-point sweep of the full range (Fig. 5). Running the same layer at inference drives it to exactly zero at 13.636 V.

Warm-starting matters as much as accuracy: starting from the network reaches a better design in 5 optimisation steps than a cold start reaches in 500.

VII. Adaptive Control Under Unmeasurable Variance

In a 2-DOF actuator the beam can tip in before it snaps down, at an angle set by a fabrication parameter that is not observable at test time. Measuring a device's ceiling destroys it, so a fixed-gain controller must target the worst-case die, and every good device is driven far below its capability.

Table VII shows the adaptive policy recovering $80.4 \pm 4.9\%$ of the available headroom, equivalent to +12.8% usable travel, with zero devices destroyed.

TABLE VIII
Measured throughput. Source: sim/bench.py.

Workload	Device	Rate
Batched design evaluation	CPU, fp64	1,430 designs/s
Amortized inference	CPU	0.04 ms/design
RL training	GPU, fp32	9.1 M env-steps/s
Lumped dynamics	GPU, fp32	179 M steps/s

TABLE IX
Every claim above traces to one script. Runtimes are laptop CPU.

Script	What it establishes	Time
test_fold.py	fold system, Λ_{PI} , c^3 law	55 s
validate_beam.py	beam vs. closed forms, grid	34 s
model2dof.py	tip-in vs. [4], 3/3 mech.	0.4 s
validate_nazemi.py	vs. 2025 measured microbridge	26 s
validate_mtest.py	vs. 1997 measured V_{PI}	24 s
sensitivity_map.py	$\xi^* = 0.217$, autodiff vs. FD	10 s
inverse_design.py	free-form design, both BCs	15 min
amortized_hard.py	design network + hard layer	26 min
analyze_rl2dof.py 3	3 RL seeds, full scale	8 min
surrogate_fair.py	MLP / FNO baselines	10 min

TABLE X
Summary. Each outcome against the baseline it is measured on.

Outcome	Baseline	Result
Drive voltage	uniform gap	22.86 $\rightarrow$ 13.60 V
Vs. published	[3] hand-tuned	13.60 vs 14.14 V
Vs. FNO surrogate	equal budget	13.60 vs 14.25 V
Training data	48,000 [1]	0
Usable travel	fixed gain	+12.8%
Devices destroyed	—	0.0%
Headroom	oracle	$80.4 \pm 4.9\%$
Turnaround	FEM campaign	15 s
Per-design	~ 30 s opt.	0.04 ms
Gradient error	Euler identity	0.000000%
Licence needed	COMSOL / ANSYS	none

The mean is not the evidence. A policy that had merely become less conservative would raise mean travel and leave the correlation near zero. What matters is whether achieved travel tracks each individual device's own ceiling: $\rho = +0.99$ against +0.06 for fixed gain (Fig. 6a). The policy beats the baseline in every ceiling bin, including the most fragile. That is inference of an unmeasurable parameter from the transient response.

VIII. Throughput and Determinism

Batched design evaluation converges on 100% of designs. The static solver requires float64, because the fourth-order stencil is cancellation-limited, and consumer GPUs run fp64 at roughly 1/64 of fp32 rate, so the solver is CPU-bound by design and no result here depends on GPU access.

All stochastic components use explicit, seeded PRNG keys, so runs are reproducible; RL results are reported across three independent seeds with standard deviations on a fixed held-out set of 512 devices.

IX. Reproducing Every Number

Table IX maps every claim to the script that produces it, and Table X collects the outcomes with the baseline each is measured against. The two notebooks linked on page 1 reproduce the fast half of this record end to end, in the browser, with nothing installed.

References

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